

Notes: Caliendo, Monte, and Rossi-Hansberg (JPE, 2015)

The Anatomy of French Production Hierarchies

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1 Big picture

- **Question.** How are firms internally organized, and how does this organization change when firms grow or shrink? The paper asks whether a firm's workforce can be usefully described as a hierarchy of layers and whether changes in layers are central to firm growth.
- **Setting.** The authors study French manufacturing firms from 2002 to 2007, using administrative balance-sheet data and employee-level occupational, wage, and hours data. The final data cover about 90% of manufacturing value added.
- **Layer idea.** Workers are grouped into hierarchical occupational layers. Blue-collar workers and clerks form the bottom production layer; supervisors, senior staff, and wage-earning owners or directors form higher layers. Firms can have up to four layers.
- **Main descriptive fact.** Firms with more layers are larger in value added and hours, and they tend to pay higher wages. Most firms have consecutively ordered layers that start at the bottom.
- **Main dynamic fact.** Firms that grow a lot tend to add layers; firms that contract tend to drop layers. The probability of adding a layer rises with value added, while the probability of dropping a layer falls with value added.
- **Key wage result.** Firms that expand without adding layers raise wages in all layers. Firms that expand by adding a layer reduce average wages in all preexisting layers. The reverse occurs when firms contract and drop a layer.
- **Mechanism.** The paper interprets wages as a measure of workers' knowledge. Adding a new upper layer lets the firm economize on knowledge in lower layers, so the firm can hire or retain less experienced workers in preexisting layers while expanding hours.
- **Economic takeaway.** Firm growth is not simply a scale-up of the same organization. Large expansions often involve reorganization, and reorganization changes the composition, wages, and knowledge structure of the firm.

2 Incremental contribution of the paper

- **First large-scale anatomy of firm hierarchies.** Earlier evidence on firm organization often came from small samples or case studies. This paper uses a broad administrative data set covering most French manufacturing value added.
- **A data-based mapping from occupations to layers.** The paper makes the theoretical concept of a production hierarchy operational by mapping French occupational categories into layers. This makes organization measurable in ordinary administrative data.
- **From firm size to organizational structure.** Standard firm-dynamics work often studies employment, investment, or productivity without observing internal organization. This paper shows that the number of layers is a meaningful state variable for firm behavior.
- **New empirical fact about reorganizing firms.** The most striking contribution is that expanding firms that add layers reduce average wages in preexisting layers. This is hard to reconcile with models in which growing firms simply hire more of the same worker types.

- **Theory-guided empirical design.** The paper is closely guided by the Caliendo and Rossi-Hansberg model of knowledge hierarchies. The empirical work tests whether the model’s moment restrictions appear in the data.
- **Layer-level rather than firm-level evidence.** The paper shows that firm-level averages can be misleading. Once the firm is decomposed by layer, the responses of hours and wages become systematic.
- **Composition margin contribution.** The paper shows that changes in layer-level wages mainly come from changing the composition of workers, not from cutting the wages of continuing workers.
- **Education-versus-experience contribution.** The paper links wage/knowledge changes to worker characteristics. Firms that grow without adding layers mainly raise education at lower layers; firms that add layers mainly lower average experience in preexisting layers.
- **A practical contribution for future research.** The paper provides a replicable way to study internal organization in any country with firm-level wages by occupation.

3 Data and measurement

3.1 Administrative data

- The paper merges two mandatory French administrative sources: BRN balance-sheet data and DADS employee reports.
- BRN provides firm balance-sheet variables such as value added and labor costs.
- DADS provides worker occupation, hours, earnings, and related employee information.
- The sample is restricted to manufacturing firms from 2002 to 2007 because the occupational categories used to construct layers changed in 2002.
- After matching and cleaning, the sample contains the large majority of French manufacturing firms and about 90% of manufacturing value added.

Interpretation: The main measurement advantage is scale. The paper can observe organizational structure for many firms, not only for a small survey sample or one case-study firm.

3.2 Construction of layers

- The authors use the PCS–ESE occupational classification for manufacturing.
- Occupation 2 contains wage-earning owners, CEOs, and firm directors.
- Occupation 3 contains senior staff and top management.
- Occupation 4 contains supervisors.
- Occupations 5 and 6 contain clerks and blue-collar workers. These are merged because their wage distributions are very similar.
- The merged clerk/blue-collar category is layer 1; supervisors are layer 2; senior staff are layer 3; CEOs/directors are layer 4.

- A firm is said to have a layer if it reports positive hours in that layer.

Interpretation: Layers are not defined by functional tasks such as accounting or production engineering. They are defined by hierarchical level and are validated using wages, hours, and transition patterns.

3.3 Firm-level and layer-level outcomes

- The main firm-level variables are value added, total hours, average hourly wages, and number of layers.
- The main layer-level variables are hours worked in each layer and average hourly wage in each layer.
- Hours are often normalized by hours in the top layer because the theory treats the top manager's time as fixed.
- Wages are interpreted as a market-based measure of knowledge or problem-solving ability.

Interpretation: The paper studies how firms adjust both the *quantity* of labor in each layer and the *knowledge/wage* of labor in each layer.

3.4 Hierarchy and consecutiveness

- Most firms have consecutively ordered layers that start from the bottom. For example, a three-layer firm usually has layers 1, 2, and 3 rather than layers 1, 3, and 4.
- Firms are hierarchical in hours: lower layers usually employ more hours than higher layers.
- Firms are hierarchical in wages: higher layers usually pay higher average wages than lower layers.
- Cross-layer wage variation explains about half of within-firm wage variation, so layers are economically meaningful.

Interpretation: The layer classification is not just a label. It captures systematic differences in firm scale, wage structure, and internal hierarchy.

3.5 Organizational transitions

- Firms can add or drop layers from one year to the next.
- Most firms keep the same number of layers in a given year, but a sizable minority reorganize.
- When firms change layers, they usually add or drop only one layer, and the change is usually consecutive.
- Firms that add layers tend to have grown faster in value added in the years before the transition.
- Firms that drop layers tend to have grown more slowly or shrunk before the transition.

Interpretation: Layer changes look like endogenous reorganizations around size thresholds, not random reclassifications of workers.

4 Theory and empirical specifications

4.1 Production hierarchy model

The theory behind the empirical analysis is a knowledge hierarchy. Production workers face problems. If they cannot solve a problem, it is passed upward to managers in higher layers. The firm chooses the number of layers, the number of employees in each layer, and the knowledge of employees in each layer.

Let $C(q, w)$ denote the minimum variable cost of producing quantity q at wage level w . The simplified cost problem is

$$C(q, w) = \min_{L, \{n_L^\ell, z_L^\ell\}_{\ell=1}^L} \sum_{\ell=1}^L n_L^\ell w_L^\ell, \quad (1)$$

subject to

$$q \leq F(Z_L^L) n_L^1, \quad (2)$$

$$w_L^\ell = w(cz_L^\ell + 1), \quad (3)$$

$$n_L^\ell = h n_L^1 [1 - F(Z_L^{\ell-1})] \quad \text{for } L \geq \ell > 1, \quad (4)$$

$$n_L^L = 1. \quad (5)$$

- L is the number of layers.
- n_L^ℓ is labor hours or employees in layer ℓ for an L -layer firm.
- z_L^ℓ is knowledge in layer ℓ .
- w_L^ℓ is the wage in layer ℓ .
- h is the communication cost of passing problems upward.

Interpretation: The firm adds layers to economize on knowledge. A new upper layer can handle rare, difficult problems, allowing lower layers to know less while the firm operates at a larger scale.

4.2 Model implications

The paper emphasizes four main implications:

- Firms are hierarchical: lower layers have more employees/hours than higher layers.
- Firms with higher demand have more layers, larger sales, and more employees.
- Conditional on the number of layers, growing firms increase both hours and wages in every layer.
- Conditional on demand, adding a layer increases employment in preexisting layers but reduces wages/knowledge in preexisting layers.

Interpretation: The model predicts two different types of growth. Growth without reorganization raises wages in existing layers. Growth with reorganization lowers wages in existing layers because the firm substitutes a new upper layer for knowledge lower in the hierarchy.

4.3 Growth without changing layers

For firms that keep the same number of layers for two consecutive years, the paper estimates how normalized hours change with value added:

$$d \ln \tilde{n}_{Lit}^\ell = \beta_L^\ell d \ln \widetilde{VA}_{it} + \varepsilon_{it}. \quad (6)$$

It also estimates how layer-level wages change with value added:

$$d \ln \tilde{w}_{Lit}^\ell = \gamma_L^\ell d \ln \widetilde{VA}_{it} + \varepsilon_{it}. \quad (7)$$

- The variables are detrended to remove aggregate time effects.
- Hours are normalized by hours in the top layer.
- The analysis is conducted separately by the number of layers and by layer.

Interpretation: These regressions describe the familiar growth margin: when a firm grows without changing its internal hierarchy, it expands hours and raises wages in existing layers.

4.4 Growth with organizational transitions

For firms that change the number of layers, the paper estimates average log changes in hours and wages for layers that exist both before and after the transition:

$$E[d \ln \tilde{n}_{Lit}^\ell \mid L \rightarrow L'] \quad (8)$$

$$E[d \ln \tilde{w}_{Lit}^\ell \mid L \rightarrow L']. \quad (9)$$

- If $L' < L$, the firm drops layers.
- If $L' > L$, the firm adds layers.
- The comparison focuses on common layers, $\ell = 1, \dots, \min\{L, L'\}$.

Interpretation: The transition analysis is the key test of reorganization. Adding layers should raise hours but lower wages in common layers; dropping layers should lower hours but raise wages in common layers.

4.5 Education, experience, and composition

The paper also asks how firms change average wages in a layer. The analysis separates two margins:

- **Extensive margin:** the firm changes the composition of workers in a layer through hiring, exits, or promotions.
- **Intensive margin:** the firm changes wages of continuing workers.

The paper then estimates worker-level education and potential labor market experience using a labor force survey and applies the estimated relationships to DADS employees.

Interpretation: The wage changes are mainly compositional. Firms that add layers reduce average experience in preexisting layers; firms that grow without adding layers rely more on education increases, especially lower in the hierarchy.

5 Identification and empirical design

5.1 What the design can identify

- The paper is not a randomized experiment and does not estimate the causal effect of an exogenous organizational shock.
- The goal is to document robust empirical patterns predicted by a theory of hierarchical organization.
- The evidence is strongest as a set of theory-guided moment restrictions: when firms add/drop layers, hours and wages move in systematic layer-specific ways.

Interpretation: The correct reading is not “adding a layer randomly causes wages to fall.” It is that firms that reorganize by adding layers systematically behave in the way predicted by hierarchy theory.

5.2 Why the layer measure is credible

- Wages are strongly ranked across occupational categories.
- Firms with more layers are larger and pay higher wages.
- Most firms have consecutive layers starting at layer 1.
- Hours and wages within firms are usually hierarchical.
- Layer transitions are related to firm size and previous growth.

Interpretation: These facts make it unlikely that layers are arbitrary titles. They behave like economically meaningful organizational units.

5.3 Main threats and limitations

- The paper cannot fully separate demand shocks, productivity shocks, and organizational choices.
- Some measured transitions may include occupational reclassification, although wage-distribution evidence argues against pure relabeling.
- French labor market institutions may shape the composition margin, so adjustment mechanisms may differ in other countries.
- The data are rich at the firm-layer level but cannot fully track the universe of individual worker careers through every transition.

Interpretation: The evidence is powerful for firm- and layer-level organization, but less suited to individual-worker welfare or fully causal policy counterfactuals.

6 Tables and figures

6.1 Figure 1: Average cost function and hierarchy

Read:

- The figure shows the lower envelope of average cost curves for organizations with different numbers of layers.
- Moving along the cost curve without adding a layer raises scale within the same hierarchy.
- Jumping to a higher-layer organization changes the wage and employment structure of all pre-existing layers.

Economic message: The figure is the theory’s visual foundation. Adding a layer is not just adding one manager; it reorganizes the entire knowledge structure of the firm.

6.2 Table 1: Distribution of wages by occupation

Read:

- Wages are clearly ordered across occupations: CEOs/directors earn the most, followed by senior staff, supervisors, and lower occupations.
- Clerks and blue-collar workers have very similar wage distributions, which motivates merging them into the bottom layer.
- The table validates wages as a one-dimensional proxy for knowledge and hierarchy.

Economic message: The occupational categories have a clear wage hierarchy, supporting the use of occupations to construct firm layers.

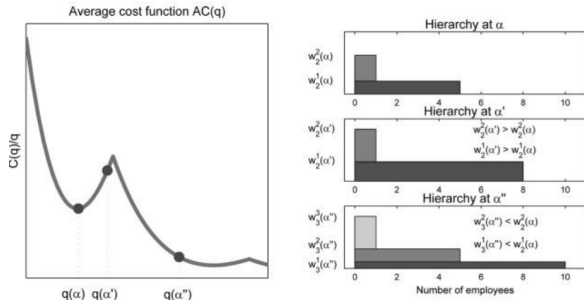


FIG. 1.—The average cost function $C(q, w/q)$ as a function of q

Figure 1: Average cost function and hierarchy

TABLE 1
DISTRIBUTION OF AVERAGE HOURLY WAGE BY OCCUPATION IN 2005 EUROS

	CEO, Directors	Senior Staff	Supervisors	Clerks	Blue-Collar
Mean	81.39	47.83	26.58	19.01	20.70
p5	23.68	21.45	14.35	10.63	10.64
p10	28.60	25.01	16.21	11.79	11.82
p25	41.51	31.00	19.36	13.84	13.65
p50	58.06	38.28	23.11	16.49	15.97
p75	80.48	47.26	27.76	19.95	19.07
p90	114.51	59.91	34.15	24.66	23.40
p95	142.29	72.08	40.45	29.37	27.87

NOTE.—This table reports, for each one-digit occupational code present in the PCS-ESE 2003, mean and percentiles of the hourly wage distribution across all firms and years in the data. One observation in an occupation is the average hourly wage in a given firm-year from the BRN source, conditional on the firm reporting the occupation. The average hourly wage is the total labor cost from the BRN data set for an occupation, divided by the number of hours reported in this occupation. The total labor cost for an occupation is computed multiplying the total labor cost from the firm balance sheet times the share of wages paid to the occupation as resulting from the DADS source. Occupation 6 excludes one outlier, which would have driven its mean to 28.89.

Table 1: Distribution of average hourly wage by occupation

6.3 Table 2: Data description by year

Read:

- The table reports the number of firms, value added, hours, wages, and average number of layers by year.
- The sample is large in every year, with more than 70,000 firms annually.
- The average number of layers is a little above 2.5 and declines slightly over the period.

Economic message: The data are broad and stable enough to study firm organization over time, not just cross-sectional differences.

6.4 Table 3: Firm characteristics by number of layers

Read:

- Firms with more layers are much larger in value added and hours.
- Median wages rise monotonically with the number of layers.
- One-layer firms are small, while four-layer firms are large organizations.

Economic message: The number of layers is strongly related to firm scale, which is exactly what hierarchy theory predicts.

TABLE 2
DATA DESCRIPTION BY YEAR

YEAR	FIRMS	AVERAGE			
		Value Added	Hours	Wage	Number of Layers
2002	78,494	2,929	78,775	22.46	2.60
2003	76,927	2,922	77,813	22.69	2.58
2004	75,555	2,957	76,574	23.47	2.59
2005	74,806	2,799	73,078	23.64	2.55
2006	73,834	2,847	72,770	23.50	2.53
2007	71,859	2,709	68,908	24.10	2.51

NOTE.—This table reports, for each year, the number of firms in the data set and corresponding averages across all firms for selected variables. Value added pertains to the firm's balance sheet. Hours is the average number of hours from the DADS source. Wage is the average hourly wage from the BRN in 2005 euros. Number of layers is the average number of layers across firms in each year. Value added is in thousands of 2005 euros.

Table 2: Data description by year

TABLE 3
DATA DESCRIPTION BY NUMBER OF LAYERS IN THE FIRM

NUMBER OF LAYERS	FIRM-YEARS	AVERAGE			
		Value Added	Hours	Wage	MEDIAN WAGE
1	80,326	201	7,656	26.90	17.50
2	124,448	401	15,706	21.82	18.64
3	160,030	2,834	80,488	22.31	20.41
4	86,671	8,916	211,098	23.89	22.04

NOTE.—This table reports summary statistics on firm-level outcomes, grouping firm-year observations according to the number of layers reported. Firm-years is the number of firm-year observations in the data with the given number of layers. Value added is the average from the firm's balance sheet. Hours is the average number of total hours from the DADS source. Wage is the average hourly wage from the BRN in 2005 euros. Median wage is the median across all firms in the cell of the average hourly wage from the BRN source in 2005 euros. Value added is in thousands of 2005 euros.

Table 3: Data description by number of layers

6.5 Figure 2: Value-added distribution by layers

Read:

- The distributions shift to the right as the number of layers increases.
- Controlling for year and industry fixed effects does not change the pattern much.
- Larger firms are much more likely to have more layers.

Economic message: More layers are associated with greater value added throughout the distribution, not only at the mean.

6.6 Figure 3: Hours distribution by layers

Read:

- Firms with more layers have many more hours worked.
- The pattern is visible in raw data and after removing year and industry fixed effects.
- The distributional evidence confirms that layers capture firm size differences.

Economic message: Hierarchical depth grows with employment scale: firms with more layers are not just higher-wage firms; they are larger production organizations.

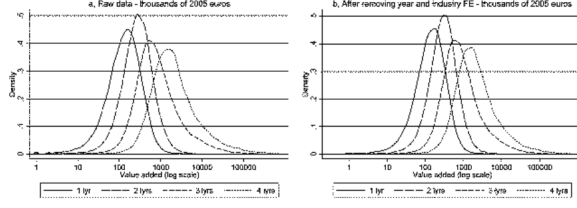


FIG. 2.—Value-added distribution by number of layers. These figures report kernel density estimates of the distribution of log value added by number of layers in the firm. The left panel reports the kernel density estimate of the distribution of log value added on the raw data: one density is estimated for each group of firms with the same number of layers. The right panel shows the same density estimated after removing year and industry fixed effects. To remove these effects, we run a regression of the form $\log v_i = \alpha + \sum \beta_j \text{layers}_{ij} + \sum \delta_j \text{industry}_j + \sum \gamma_j \text{year}_j + \varepsilon_i$, where v_i is value added for firm-year i , and layers_{ij} , industry_j , and year_j are a set of layers, two-digit industry, and year dummies, respectively. The omitted dummy for layers is for firms with zero layers of management. The log value added for firm-year i without year and industry fixed effects is then $\log \hat{v}_i = \log m + \sum \beta_j \text{layers}_{ij} + \hat{\varepsilon}_i$, where we set m to the median value added in 2002 for firms with zero layers of management. We then compute four kernel density estimates of the distribution of $\log \hat{v}_i$, grouping firms according to their number of layers.

Figure 2: Value-added distribution by number of layers

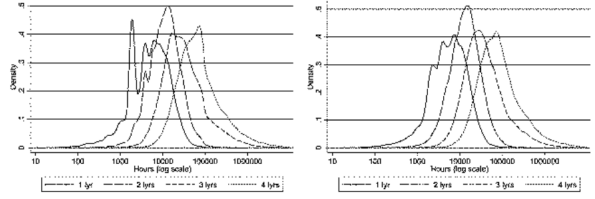


FIG. 3.—Hours distribution by number of layers. These figures report kernel density estimates of the distribution of log hours worked by number of layers in the firm. See the note to figure 2 for a description of how the densities are computed.

Figure 3: Hours distribution by number of layers

6.7 Figure 4: Average wage distribution by layers

Read:

- Average wages tend to be higher in firms with more layers.
- The wage distributions overlap, but the rightward shift is clear, especially after fixed-effect adjustment.
- This complements the occupation-level wage ranking in Table 1.

Economic message: Deeper firms are larger and tend to pay more, but the paper later shows that this cross-sectional fact can coexist with wage declines during layer-adding reorganizations.

6.8 Table 4: Consecutively ordered layers

Read:

- Most firms have layers ordered consecutively from the bottom.
- The share is even larger when weighted by value added or hours.
- Nonconsecutive layer structures are mostly small and economically less important.

Economic message: Firms usually build hierarchies from the production layer upward, which supports treating layers as a vertical organizational structure.

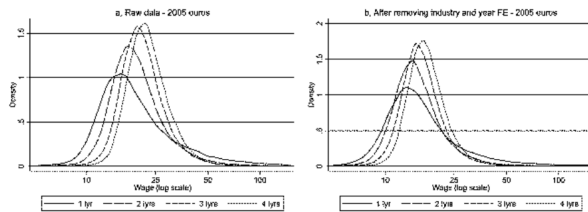


FIG. 4.—Firm average hourly wage distribution by number of layers. These figures report kernel density estimates of the distribution of log hourly wage by number of layers in the firm. See the note to figure 2 for a description of how the densities are computed.

Figure 4: Firm average hourly wage distribution by number of layers

TABLE 4 PERCENTAGE OF FIRMS THAT HAVE CONSECUTIVELY ORDERED LAYERS				
	AMONG FIRMS WITH			
	1 Layer	2 Layers	3 Layers	4 Layers
Unweighted	87.42	67.39	80.01	100
Weighted by value added	87.69	68.40	94.60	100
Weighted by hours	99.17	72.56	93.07	100

NOTE.—This table reports the fraction of firms with consecutively ordered layers conditioning on the number of layers in the firm (first four columns) and overall (fifth column). The first row reports the simple fraction of firms; the second and third rows assign a weight that is proportional to the total value added in the balance sheet and to the total hours in the DADS, respectively.

Table 4: Percentage of firms with consecutively ordered layers

6.9 Table 5: Hierarchy in hours

Read:

- Many firms employ more hours in lower layers than in higher layers.
- The hierarchy is strongest for two-layer firms and weaker when all layers must satisfy the ordering simultaneously.
- Individual adjacent-layer comparisons are hierarchical in most cases.

Economic message: The representative firm has a pyramid-like hours structure: many production workers and fewer higher-level managers.

6.10 Table 6: Hierarchy in wages

Read:

- Wages are typically higher in upper layers than in lower layers.
- The wage hierarchy is very strong in adjacent comparisons.
- The table reinforces that layers are ordered not only by title but also by compensation.

Economic message: The hierarchy is both quantitative and qualitative: lower layers contain more hours, while upper layers contain higher-wage workers.

TABLE 5
FIRMS THAT SATISFY A HIERARCHY IN HOURS

Number of Layers	$N_\ell^1 \geq N_{\ell+1}^1$ All ℓ	$N_\ell^1 \geq N_\ell^2$	$N_\ell^2 \geq N_\ell^3$	$N_\ell^3 \geq N_\ell^4$
2	85.6	85.6	...	...
3	63.4	85.9	74.8	...
4	56.5	86.9	77.5	86.9

NOTE.—This table reports, among all firms with $L = 2, 3, 4$ layers, the fraction of firms that satisfy a hierarchy in hours at all layers (first column) and the fraction of those that satisfy a hierarchy in hours between layer ℓ and $\ell + 1$, with $\ell = 1, \dots, L - 1$ (second to fourth columns). A firm satisfies a hierarchy in hours between layers number ℓ and $\ell + 1$ in a given year if the number of hours worked in layer ℓ is at least as large as the number of hours worked in layer $\ell + 1$; moreover, a firm satisfies a hierarchy at all layers if the number of hours worked in layer ℓ is at least as large as the number of hours in layer $\ell + 1$, for all layers in the firm. The term N_ℓ^i is the number of hours reported in layer ℓ in a firm with L layers from the DADS source.

Table 5: Firms that satisfy a hierarchy in hours

TABLE 6
FIRMS THAT SATISFY A HIERARCHY IN WAGES

Number of Layers	$w_\ell^{i+1} \geq w_\ell^i$ All ℓ	$w_\ell^1 \geq w_\ell^2$	$w_\ell^2 \geq w_\ell^3$	$w_\ell^3 \geq w_\ell^4$
2	85.60	85.60	...	...
3	63.40	85.90	74.80	...
4	56.50	86.90	77.50	86.90

NOTE.—This table is the same as table 5 for the case of wages, where w_ℓ^i is the average hourly wage in layer ℓ from the BRN in an L -layer firm.

Table 6: Firms that satisfy a hierarchy in wages

6.11 Figure 5: Representative hierarchies

Read:

- Each rectangle represents a layer, with width proportional to normalized hours and height proportional to wage.
- Lower layers are wider, while upper layers are taller.
- Firms with more layers have larger bottom layers and richer internal structure.

Economic message: The figure makes the organizational pyramid concrete: hierarchy is visible in both hours and wages.

6.12 Table 7: Cross-layer wage variation

Read:

- About half of within-firm wage variation is explained by cross-layer variation.

- The explanatory share rises with the number of layers.
- For four-layer firms, layers explain roughly two-thirds of wage variation in the unweighted statistic.

Economic message: Layers are important for understanding wage inequality inside firms. Ignoring layers loses a major source of wage variation.

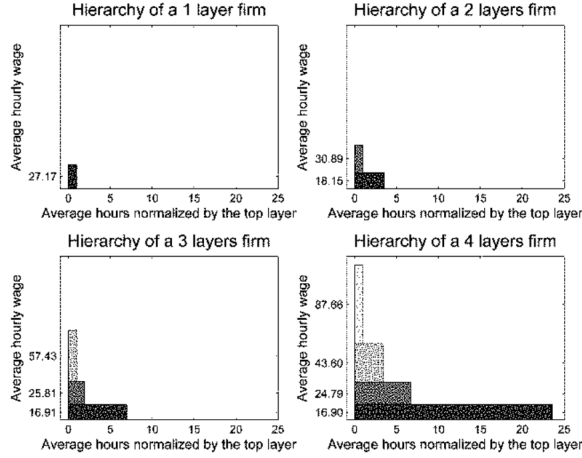


FIG. 5.—Representative hierarchies normalized by hours in the top layer. These figures portray the representative hierarchy of an L -layer firm. We first consider all firms with a given number of layers $L = 1, 2, 3, 4$ and focus on the middle tercile according to total value added. For each layer $\ell = 1, \dots, L$, we then compute the average number of hours and the average hourly wage in the layer from the BRN in this group. In a figure portraying a firm with L layers, L rectangles are shown, one for each layer. The horizontal length of each rectangle is proportional to the number of normalized hours in the layer (hours are reported along the x-axis) while its height is proportional to the average hourly wage (the value of the wage is reported to the left of each rectangle).

Figure 5: Representative hierarchies normalized by top-layer hours

TABLE 7
MEAN SHARE OF VARIATION IN WAGES EXPLAINED BY CROSS-LAYER VARIATION

	FIRM-YEARS	UNWEIGHTED	WEIGHTED BY	
			HOURS	Value Added
All firms	434,872	.50	.51	.49
Firms with more than 1 layer	370,997	.59	.51	.50
Firms with 1 layer	63,875	.00	.00	.00
Firms with 2 layers	124,299	.50	.41	.43
Firms with 3 layers	160,028	.62	.51	.50
Firms with 4 layers	86,670	.66	.53	.50

NOTE.—For this table we compute the R^2 of a regression of log hourly wages of workers within a firm on a constant and dummies for layers (all except one), weighted by the number of hours each worker provides to the firm. For each row, the column unweighted reports the average R^2 across all firm-years, while the remaining two columns to the right report the same average when weighting firms by their total number of hours or total value added. The column firm-years reports the number of firm-years used to compute the statistics in the corresponding row. Note that for some firms—e.g., firms with only one worker—the R^2 cannot be computed, and hence the total number of firm-years in the data set does not correspond to the total number of firm-years used. Each row differs from the others according to the subsample of firm-years used in computing the average.

Table 7: Mean share of wage variation explained by cross-layer variation

6.13 Table 8: Distribution of layer transitions

Read:

- Most firms keep the same number of layers from year to year.
- When firms change layers, they usually add or drop only one layer.
- Exit rates are lower for firms with more initial layers.

Economic message: Organizational change is not constant churn. Layer transitions are discrete and usually incremental.

6.14 Figure 6: Transition probabilities and value added

Read:

- The probability of adding layers increases with value added.
- The probability of dropping layers decreases with value added.
- Transition probabilities are ordered in the way the theory predicts.

Economic message: Layer changes are related to firm scale. Larger firms are closer to adding managerial layers; smaller firms are closer to dropping them.

TABLE 8
DISTRIBUTION OF LAYERS AT $t + 1$ CONDITIONAL ON LAYERS AT t

NUMBER OF LAYERS AT t	NUMBER OF LAYERS AT $t + 1$					TOTAL
	Exit	1	2	3	4	
1	15.3	67.5	15.2	1.9	.2	100
2	9.8	10.7	62.2	16.2	1.1	100
3	7.7	1.2	13.1	67.6	10.5	100
4	6.2	.2	2.0	20.5	71.3	100

NOTE.—This table reports the distribution of the number of layers at time $t + 1$, grouping firms according to the number of layers at time t . Among all firms with L layers ($L = 1, \dots, 4$) in any year from 2002 to 2006, the columns report the fraction of firms that have layers 1, . . . , 4 the following year (from 2003 to 2007) or are not present in the data set, exit. The elements in the table sum to 100 percent by row.

Table 8: Distribution of layers at $t + 1$ conditional on layers at t

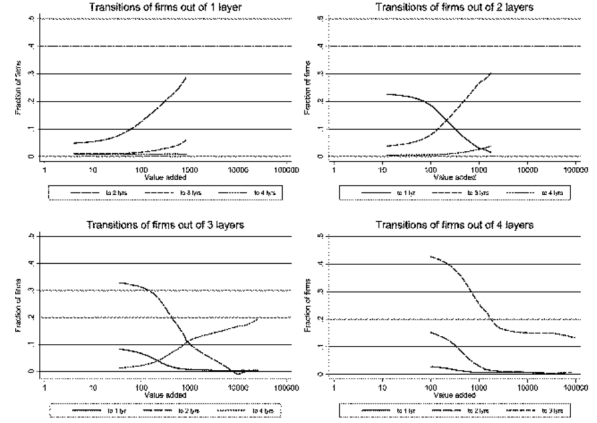


FIG. 6.—Transitions across layers depend on value added. These figures show the probability of transition away from the current layer as a function of the initial value added of the firm. Each panel reports transition probabilities starting from a different initial number of layers. To produce the panel of transitions out of layer $L = 1, \dots, 4$, we take for each year (from 2002 to 2006) all the firms with L layers and group them into 100 bins according to their value added; for each bin, we compute the fraction of firms that will have any number of layers (or exit the data set) in the following period and plot the average value added in the bin against this fraction. For each transition series we then apply a loess smoothing for all the probabilities estimated from the first to the ninety-ninth bin.

Figure 6: Transitions across layers depend on value added

6.15 Figure 7: Value added before and during transitions

Read:

- Firms that add layers grow faster before the transition.
- Firms that drop layers grow more slowly or shrink before the transition.
- The size of the transition is related to the strength of the growth or contraction pattern.

Economic message: Reorganization appears to follow accumulated growth or decline, consistent with size thresholds for changing the number of layers.

6.16 Table 9: Elasticity of hours for firms that do not change layers

Read:

- Conditional on keeping the same number of layers, growing firms increase normalized hours in lower layers.
- The elasticity is positive and significant for most entries.
- Lower layers tend to expand proportionally more than higher layers.

Economic message: When firms grow without reorganizing, they become flatter: the base of the hierarchy expands relative to the top.

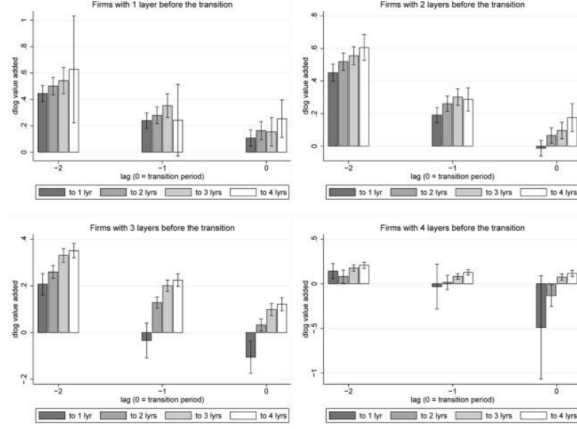


FIG. 7.—Changes in value added before and during transition. These figures show the fixed effect of a given transition (from L to L' number of layers, with $L, L' = 1, 2, 3, 4$) on the average firm growth in a period, conditional on initial size. We use the specification presented in the text. The figure presents the value of $\gamma_{L'L-k}^1$ and its 95 percent confidence interval, grouping in each panel firms with a given number of layers before the transition (i.e., the top-left panel shows $\gamma_{1'L-k}^1$, the top-right $\gamma_{2'L-k}^1$, and so on). Within each panel, the dummies are grouped according to the time lag before the transition, $k = 2, 1, 0$, and, within each group, ordered by the number of layers L' after the transition.

Figure 7: Changes in value added before and during transition

TABLE 9
ELASTICITY OF HOURS WITH VALUE ADDED FOR FIRMS THAT DO NOT CHANGE L

Number of Layers	Layer	β_ℓ^1	Standard Error	p -Value	Observations
2	1	.042	.012	.000	64,536
3	1	.039	.009	.000	91,253
3	2	.013	.010	.200	91,253
4	1	.107	.014	.000	52,799
4	2	.051	.013	.000	52,799
4	3	.037	.013	.000	52,799

NOTE.—This table reports the results of regressions of detrended log change in normalized hours at layer ℓ in a firm with L layers on its detrended log change in value added, and no constant, selecting all the firms that stay at L layers across 2 consecutive years. The term β_ℓ^1 is the coefficient on log change in value added.

Table 9: Elasticity of hours with value added when L does not change

6.17 Table 10: Elasticity of wages for firms that do not change layers

Read:

- Conditional on keeping the same number of layers, growing firms raise wages in all layers.
- Wage elasticities are positive and statistically significant throughout the table.
- Wage growth tends to be larger in higher layers.

Economic message: Without reorganization, firm growth looks conventional: larger scale is associated with more labor and higher wages/knowledge in existing layers.

6.18 Table 11: Firm-level outcomes around layer changes

Read:

- Firms that add layers increase total hours, normalized hours, and value added.
- Average wages in common layers fall sharply when firms add layers.
- Firms that drop layers show the opposite pattern: common-layer wages rise.
- Firms that change layers account for a large share of value-added changes.

Economic message: Reorganizing firms are not a fringe phenomenon. They are central to aggregate firm dynamics and wage changes.

TABLE 10
ELASTICITY OF WAGES WITH VALUE ADDED FOR FIRMS THAT DO NOT CHANGE L

Number of Layers	Layer	γ_L^e	Standard Error	p -Value	Observations
1	1	.077	.007	.000	45,045
2	1	.100	.006	.000	64,536
2	2	.118	.006	.000	64,536
3	1	.145	.006	.000	91,253
3	2	.155	.006	.000	91,253
3	3	.170	.006	.000	91,253
4	1	.171	.009	.000	52,799
4	2	.185	.009	.000	52,799
4	3	.186	.010	.000	52,799
4	4	.217	.011	.000	52,799

NOTE.—This table reports the results of regressions of log change in hourly wage by layer on log change in value added for firms that do not change their number of layers L across two consecutive periods, where both variables are detrended as specified in the main text. Specifically, we run a regression of detrended log change in average hourly wage at layer ℓ in a firm with L layers on the detrended log change in value added across all the firms that stay at L layers across two consecutive years, with no constant. The term γ_L^e is the coefficient on log change in value added. We use hourly wage at layer ℓ from the BRN.

Table 10: Elasticity of wages with value added when L does not change

TABLE 11
CHANGE IN FIRM-LEVEL OUTCOMES

	All	Increase L	No Change in L	Decrease L
$d \ln$ total hours	-.015***	.040***	-.012***	-.081***
Detrended	. . .	.055***	.003***	-.066***
$d \ln$ normalized hours	-.011***	1.362***	.012***	-1.404***
Detrended	. . .	1.373***	.023***	-1.392***
$d \ln$ VA	-.008***	.032***	-.007***	-.050***
Detrended	. . .	.040***	.001	-.041***
$d \ln$ average wage	.019***	.015***	.019***	.025***
Detrended	. . .	-.005***	.000	.006***
Common layers	.021***	-.101***	.019***	.143***
Detrended	. . .	-.122***	-.002***	.122***
% of firms	100	12.65	73.66	13.68
% value added change	100	40.12	65.08	-5.19

NOTE.—This table reports changes in firm-level outcomes between consecutive years for all firms and for the subsets of those that increase, do not change, and decrease layers. It reports changes in log hours, log normalized hours, log average wage from the BRN, and log average wage in common layers for the whole sample. The change in average wage for common layers in a firm that transitions from L to L' layers is the change in the average wage from the BRN computed using only the first $\min\{L, L'\}$ layers before and after the transition. To detrend a variable, we subtract from all the log changes in a given year the average change during the year across all firms. In the last two rows of the table, % of firms is the percentage of firms observed having each type of behavior; % value added change is the fraction of the total change in real value added observed in the data set accounted for by firms making the given transition.

*** Significant at 1 percent.

Table 11: Change in firm-level outcomes

6.19 Table 12: Hours changes for transitioning firms

Read:

- Firms that add layers increase normalized hours in common layers.
- Firms that drop layers reduce normalized hours in common layers.
- The results hold layer by layer, not just at the firm level.

Economic message: Reorganization accompanies real expansion or contraction of the hierarchy, not just relabeling of job titles.

6.20 Table 13: Wage changes for transitioning firms

Read:

- Firms that add layers reduce wages in all preexisting layers.
- Firms that drop layers raise wages in the remaining common layers.
- Upper preexisting layers often experience larger wage changes.

Economic message: This is the core novel fact of the paper. Large expansions that add layers lower average wages/knowledge in old layers because the new top layer changes the allocation of problem solving.

TABLE 12
AVERAGE LOG CHANGE IN HOURS FOR FIRMS THAT TRANSITION

Number of Layers Before	Number of Layers After	Layer	$d\ln\tilde{w}_{\ell,t}$	Standard Error	p-Value	Observations
1	2	1	1.537	.018	.000	10,177
1	3	1	1.762	.056	.000	1,263
1	4	1	2.266	.212	.000	97
2	1	1	-1.582	.017	.000	11,106
2	3	1	.716	.012	.000	16,800
2	3	2	.539	.012	.000	16,800
2	4	1	1.205	.049	.000	1,129
2	4	2	1.004	.048	.000	1,129
3	1	1	-1.795	.048	.000	1,584
3	2	1	-.682	.012	.000	17,666
3	2	2	-.518	.012	.000	17,666
3	4	1	1.352	.014	.000	14,113
3	4	2	1.289	.016	.000	14,113
3	4	3	1.174	.016	.000	14,113
4	1	1	-2.119	.173	.000	123
4	2	1	-1.059	.041	.000	1,456
4	2	2	-.918	.040	.000	1,456
4	3	1	-1.411	.014	.000	15,160
4	3	2	-1.345	.015	.000	15,160
4	3	3	-1.260	.015	.000	15,160

NOTE.—This table reports estimates of the average detrended log change in normalized hours at each layer among firms that transition from L to L' layers, with $L \neq L'$: for a transition from L to L' , we can evaluate only changes for layer number $\ell = 1, \dots, \min\{L, L'\}$. The detrending is explained in the main text. The term $d\ln\tilde{w}_{\ell,t}$ is the average detrended log change in the transition, estimated as a regression of the detrended log change in the number of normalized hours in layer ℓ in 2 consecutive years on a constant. The table uses all observed transitions in the sample.

Table 12: Average log change in hours for firms that transition

TABLE 13
AVERAGE LOG CHANGE IN WAGES FOR FIRMS THAT TRANSITION

Number of Layers Before	Number of Layers After	Layer	$d\ln\tilde{w}_{\ell,t}$	Standard Error	p-Value	Observations
1	2	1	-.129	.005	.000	10,177
1	3	1	-.332	.020	.000	1,263
1	4	1	-.678	.117	.000	97
2	1	1	.167	.005	.000	11,106
2	3	1	-.050	.002	.000	16,800
2	3	2	-.255	.004	.000	16,800
2	4	1	-.150	.015	.000	1,129
2	4	2	-.409	.019	.000	1,129
3	1	1	.356	.018	.000	1,584
3	2	1	.059	.002	.000	17,666
3	2	2	.249	.004	.000	17,666
3	4	1	-.021	.002	.000	14,113
3	4	2	-.067	.003	.000	14,113
3	4	3	-.199	.004	.000	14,113
4	1	1	.804	.109	.000	123
4	2	1	.139	.012	.000	1,456
4	2	2	.372	.016	.000	1,456
4	3	1	.009	.002	.000	15,160
4	3	2	.040	.003	.000	15,160
4	3	3	.134	.004	.000	15,160

NOTE.—This table reports estimates of the average detrended log change in hourly wage at each layer ℓ among firms that transition from L to L' layers, with $L \neq L'$: for a transition from L to L' , we can evaluate only changes for layer number $\ell = 1, \dots, \min\{L, L'\}$. The term $d\ln\tilde{w}_{\ell,t}$ is the average detrended log change in the transition, estimated as a regression of the detrended log change in the average hourly wage at layer ℓ in 2 consecutive years on a constant. The table uses all observed transitions in the sample.

Table 13: Average log change in wages for firms that transition

6.21 Table 14: Decomposition of wage changes

Read:

- Workers in common layers earn less after firms add layers.
- The newly added upper layer earns more than the firm's previous average wage.
- The total average wage effect combines lower common-layer wages with high pay in the new top layer.

Economic message: Adding a layer can raise the wage bill at the top while lowering wages in preexisting layers. Firm-level average wages alone can hide this reorganization.

6.22 Figure 8: Wage distribution before and after transitions

Read:

- The wage distribution changes significantly after layer transitions.
- When firms add layers, lower parts of the distribution tend to shift downward.
- When firms drop layers, the distribution tends to shift upward.

Economic message: The results are not merely mechanical averages. Reorganization shifts the wage distribution within firms.

$$d \ln \bar{w}_{t,\ell} = \ln \bar{w}_{t,\ell'} - \ln \bar{w}_{t,\ell} \\ = \ln[(\bar{w}_{t,\ell'}^{sL}/\bar{w}_{t,\ell})s + (\bar{w}_{t,\ell'}^{L'}/\bar{w}_{t,\ell})(1-s)],$$

TABLE 14
DECOMPOSITION OF TOTAL LOG CHANGE IN AVERAGE WAGES

From/To	$\bar{w}_{t,\ell'}^{sL}/\bar{w}_{t,\ell}$			From/To	$\bar{w}_{t,\ell'}^{L'}/\bar{w}_{t,\ell}$		
	2	3	4		2	3	4
1	.963*** (10,167)	.865*** (1,262)	.733*** (96)	1	1.507*** (10,166)	1.501*** (1,263)	1.602*** (97)
2		.926*** (16,783)	.876*** (1,128)	2		2.040*** (16,783)	2.021*** (1,129)
3			.958*** (14,099)	3			4.385*** (14,099)
From/To	s			From/To	$d \ln \bar{w}_{t,\ell}$		
	2	3	4		2	3	4
1	.741*** (10,166)	.620*** (1,262)	.563*** (97)	1	-.007*	-.094***	-.305**
2		.853*** (16,784)	.775*** (1,128)	2		.005**	-.033**
3			.948*** (14,099)	3			-.001 (14,098)

NOTE.—This table reports the sources of change in the average hourly wage from the BRN, by type of transition. For a given firm transitioning from L to $L' > L$, write the detrended log change in average wage as

$$d \ln \bar{w}_{t,\ell} = \ln \bar{w}_{t,\ell'} - \ln \bar{w}_{t,\ell} = \ln[(\bar{w}_{t,\ell'}^{sL}/\bar{w}_{t,\ell})s + (\bar{w}_{t,\ell'}^{L'}/\bar{w}_{t,\ell})(1-s)].$$

In this notation, $\bar{w}_{t,\ell'}^{sL}$ is the average hourly wage after the transition in the common layers, $\bar{w}_{t,\ell}$ is the average hourly wage before the transition, s is the share of hours of the common layers after the transition, and $\bar{w}_{t,\ell'}^{L'}$ is the average wage in the layers added after the transition. We report in the cells the average of each of these quantities in the first three panels; the fourth panel shows the overall average log change in hourly wage during the indicated transition. Each cell is computed excluding observations below the 0.05th and above the 99.95th percentile. Numbers of observations are in parentheses.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Table 14: Decomposition of total log change in average wages

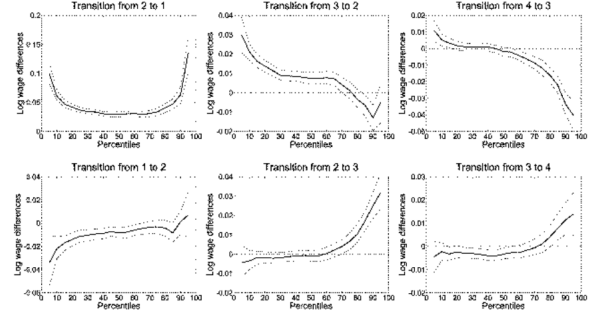


FIG. 8.—Difference in the distribution of wages after minus before the transition. These figures portray, for each transition type, difference in the distribution of wages after minus before the transition. The axis in each panel reports the log difference between the wage at the p th percentile after, less its correspondent wage before the transition. To compute standard errors, we performed 500 bootstrap replications of this process, clustering the sampling at the firm-transition level (i.e., one cluster contains all the employees present either before or after a transition in one firm) to preserve the within-firm and within-transition correlation in wages present in the data, and we report the 5th and 95th percentiles of these replications. To compute these panels, we first construct an employee-level data set that contains log hourly wage of each employee, a firm identifier, and year and current number of layers of the firm, and we remove year and firm fixed effects from the log hourly wage distribution. We then focus on all the hours worked in firm-year observations in which the firm is making a transition from ℓ to ℓ' layers (both before and after the transition takes place) and compute the p th percentile (for $p = 5, 10, \dots, 95$) in the two distributions: the wage distribution after and before the transition. Each distribution of hourly wages is computed making sure that each employee receives a weight proportional to her number of hours in the firm but giving to each firm the same weight, regardless of the total number of hours worked in it (to be consistent with our estimates of changes in log wages in table 13).

Figure 8: Difference in wage distribution after minus before transition

6.23 Figure 9: Wage distribution in common layers

Read:

- The figure removes the newly added or dropped layer and focuses only on common layers.
- When firms add layers, the common-layer wage distribution shifts downward across percentiles.
- When firms drop layers, the common-layer wage distribution shifts upward.

Economic message: This addresses the relabeling concern. Even among layers that exist before and after the transition, wages shift in the predicted direction.

6.24 Figure 10: Growing firms that do not transition

Read:

- For firms that grow but do not change layers, the wage distribution shifts upward.
- This pattern holds across firms with different numbers of layers.
- It contrasts sharply with the downward shift in common layers for firms that add layers.

Economic message: The distinction between growth with reorganization and growth without reorganization is essential. The same value-added growth can imply opposite wage movements depending on layer changes.

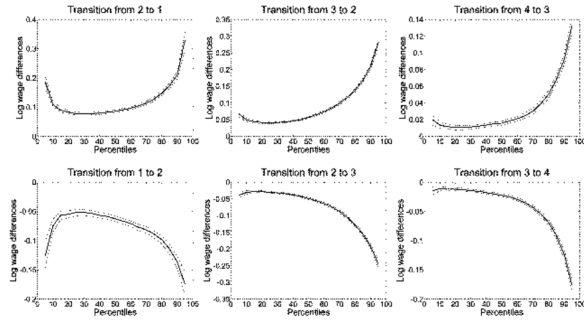


FIG. 9.—Difference in the distribution of wages in common layers after minus before the transition. This figure portrays, for each transition type, the estimated log change (on the y-axis) in the percentiles (on the x-axis) of the wage distribution in common layers within firms, after the transition versus before the transition, and associated 95 percent bootstrapped confidence intervals. To build it we follow the same process described in figure 8, to which we refer, with the only difference that after removing year and firm fixed effects, we focus on the wage distribution implied by all hours worked in layers that are common before and after the transition.

Figure 9: Difference in wage distribution in common layers

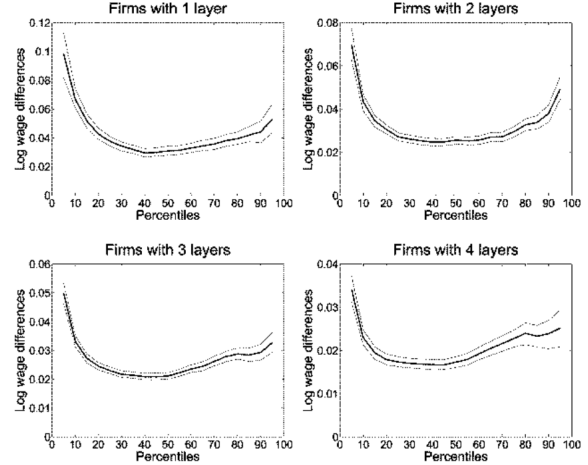


FIG. 10.—Difference in the distribution of wages for firms that do not transition and $d \ln VA > 0$. This figure portrays, for firms staying at a given number of layers in 2 consecutive years and positive change in value added, the estimated log change (on the y-axis) in the percentiles (on the x-axis) of the wage distribution within firms, the year after less the year before, and associated 95 percent bootstrapped confidence intervals. To build it we follow the same process described in figure 8.

Figure 10: Wage distribution for growing firms that do not transition

6.25 Figure 11: Representative transitions between one and two layers

Read:

- The top panels show a one-layer firm adding a second layer.
- The bottom panels show a two-layer firm dropping to one layer.
- The figure illustrates large changes in normalized hours and wages around reorganization.

Economic message: Even the simplest hierarchy transition changes the shape of the firm. Adding a layer creates a managerial level and expands the base.

6.26 Figure 12: Representative transitions between two and three layers

Read:

- A two-to-three-layer transition adds a new top layer and expands lower layers.
- A three-to-two-layer transition contracts the hierarchy and raises wages in remaining layers.
- The visual pattern mirrors the table results on hours and wages.

Economic message: Medium-depth firms reorganize in the same systematic way: expansions add layers and lower common-layer wages; contractions remove layers and raise common-layer wages.

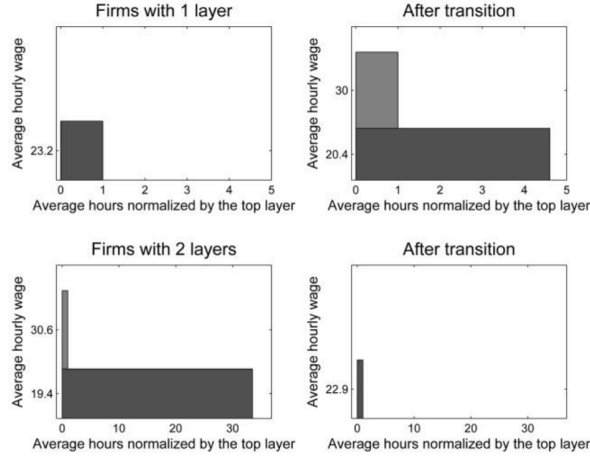


FIG. 11.—Representative transitions between firms with one and two layers. These figures depict hierarchies of firms that transition from L to $L + 1$ layers and from $L + 1$ to L . In each panel, for fixed L , the left column portrays the representative hierarchy with L layers and normalized hours, using only firms that will make the indicated transition. Each representative hierarchy is computed as in figure 5. The right column estimates the hierarchy after the transition using the average log changes in each quantity as resulting from our estimates reported in tables 12 and 13: the wage (normalized hours) in layer ℓ after the transition from L to $L + 1$ is computed as the average hourly wage (normalized hours) before the transition multiplied times $\exp(b)$, where b is the average log change in wages (normalized hours) at layer ℓ in transitions from L to $L + 1$. Normalized hours after a transition are always set to one for the top layer. An analogous procedure is followed for transitions from $L + 1$ to L . For transitions one layer up, the hourly wage for the top layer after the transition is computed as follows: we estimate the average log change in the detrended wage of the top layer and multiply the wage at the top layer before the transition times the exponential of this change.

Figure 11: Representative transitions between one and two layers

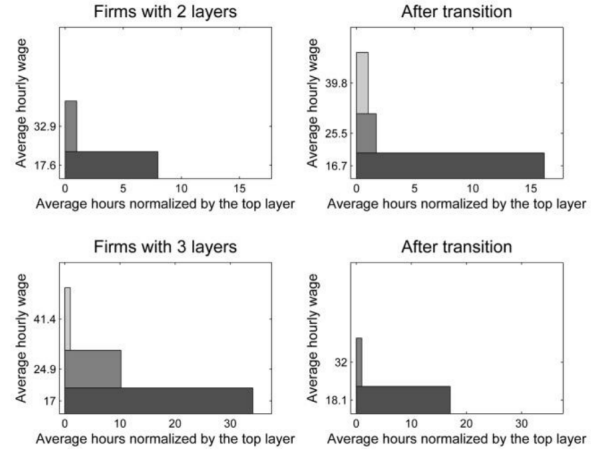


FIG. 12.—Representative transitions between firms with two and three layers. These figures depict hierarchies of firms that transition from two to three layers and from three to two. To build it we follow the same process described in figure 11.

Figure 12: Representative transitions between two and three layers

6.27 Figure 13: Representative transitions between three and four layers

Read:

- The figure shows the most complex transition in the data: adding or dropping a fourth layer.
- When firms add the fourth layer, they expand lower layers and introduce a high-wage top layer.
- When firms drop the fourth layer, the remaining hierarchy becomes smaller and wages in common layers rise.

Economic message: The theory scales up to deep hierarchies. Reorganization affects every layer, not only the newly added or removed one.

6.28 Table 15: Experience and education for firms that do not change layers

Read:

- For firms that grow without changing layers, education tends to rise, especially in lower layers.
- Experience effects are more mixed, with positive effects more visible at higher layers.
- The table decomposes wage/knowledge changes into worker-characteristic changes.

Economic message: Non-reorganizing growth raises knowledge partly by hiring more educated workers, particularly in the lower part of the hierarchy.

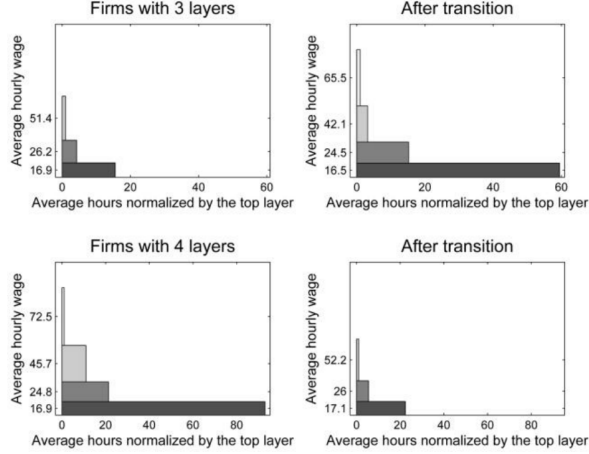


FIG. 13.—Representative transitions between firms with three and four layers. These figures depict hierarchies of firms that transition from three to four layers and from four to three. To build it we follow the same process described in figure 11.

Figure 13: Representative transitions between three and four layers

TABLE 15
ELASTICITY OF KNOWLEDGE WITH VALUE ADDED FOR FIRMS THAT DO NOT CHANGE L

Number of Layers	Layer	Experience	p -Value	Education	p -Value	Observations
1	1	.001	.690	.002	.030	45,009
2	1	-.010	.010	.004	.000	64,469
2	2	.009	.030	.003	.000	64,469
3	1	-.010	.000	.004	.000	91,161
3	2	-.001	.970	.003	.000	91,161
3	3	.008	.000	.001	.100	91,161
4	1	-.015	.000	.003	.000	52,730
4	2	-.004	.280	.003	.000	52,730
4	3	.000	.970	.000	.790	52,730
4	4	.007	.020	-.003	.070	52,730

NOTE.—This table reports the results of regressions of log change in years of potential labor market experience and of education by layer on log change in value added for firms that do not change their number of layers L across two consecutive periods, where both variables are detrended as specified in the main text. Specifically, we run a regression of detrended log change in each of the two variables at layer ℓ in a firm with L layers on its detrended log change in value added, and no constant, across all the firms that stay at L layers across 2 consecutive years. The columns p -value report the respective p -value for each left-hand side using robust standard errors. We use specification 2 to estimate these measures (for a description of the specifications, refer to the online app.).

Table 15: Elasticity of knowledge with value added when L does not change

6.29 Table 16: Experience and education for firms that change layers

Read:

- Firms that add layers tend to reduce average experience in preexisting layers.
- Firms that drop layers tend to increase average experience in common layers.
- Education changes are smaller and less systematic than experience changes.

Economic message: Reorganization changes the knowledge structure mainly through experience. When a new top layer is added, lower layers can be staffed with less experienced workers.

TABLE 16
AVERAGE CHANGE IN KNOWLEDGE FOR FIRMS THAT CHANGE L

Number of Layers Before	Number of Layers After	Layer	Experience	p -Value	Education	p -Value	Observations
1	2	1	-.108	.000	-.004	.000	10,171
1	3	1	-.184	.000	-.003	.290	1,261
1	4	1	-.330	.000	.025	.030	97
2	1	1	.096	.000	.005	.000	11,088
2	3	1	-.044	.000	.000	.820	16,778
2	3	2	-.181	.000	.002	.010	16,778
2	4	1	-.064	.000	.002	.290	1,124
2	4	2	-.228	.000	.008	.010	1,124
3	1	1	.137	.000	.006	.000	1,584
3	2	1	.044	.000	.002	.530	17,626
3	2	2	.153	.000	.000	.000	17,626
3	4	1	-.011	.000	.001	.100	14,098
3	4	2	-.038	.000	-.001	.000	14,098
3	4	3	-.176	.000	.024	.820	14,098
4	1	1	.197	.000	-.002	.950	123
4	2	1	.073	.000	.000	.120	1,454
4	2	2	.172	.000	-.005	.000	1,454
4	3	1	.013	.000	-.002	.260	15,150
4	3	2	.025	.000	-.001	.000	15,150
4	3	3	.113	.000	-.020	.000	15,150

NOTE.—This table reports estimates of the average detrended log change in years of potential labor market experience and of education at each layer ℓ among firms that transition from L to L' layers, with $L \neq L'$: for a transition from L to L' , we can evaluate only changes for layer number $\ell = 1, \dots, \min\{L, L'\}$. The detrending is explained in the main text. Each average change is estimated as a regression of the detrended log change in the variable of interest in layer ℓ in 2 consecutive years on a constant. We use specification 2 to estimate these measures (for a description of the specifications, refer to online app. B).

Table 16: Average change in knowledge for firms that change L

7 Short summary

- The paper studies French manufacturing firms from 2002 to 2007 using administrative data on balance sheets, occupations, wages, and hours.
- It constructs up to four layers from occupational categories: production workers/clerks, supervisors, senior staff, and wage-earning owners or directors.
- Firms with more layers are larger, employ more hours, and tend to pay higher wages.
- Most firms have consecutive layers that start from the bottom, and most firms are hierarchical in both hours and wages.
- The probability of adding a layer rises with value added; the probability of dropping a layer falls with value added.
- Firms that grow without changing layers increase hours and wages in all layers.
- Firms that grow by adding layers increase hours but reduce average wages in all preexisting layers.
- Firms that contract and drop layers show the reverse pattern: hours fall and wages in common layers rise.
- These wage changes are not just relabeling; wage distributions in common layers shift in the predicted direction.

- The wage/knowledge adjustment mainly occurs through worker composition, especially changes in experience.
- The main contribution is to show that reorganization through layers is central to firm growth and wage dynamics.

8 Conclusion

8.1 Core conclusion

Caliendo, Monte, and Rossi-Hansberg show that the internal organization of firms can be measured in large administrative data and that this organization changes systematically as firms grow or shrink. The central empirical result is that layer changes are essential to understanding firm dynamics. Firms that grow without changing layers raise wages in existing layers, but firms that grow by adding layers lower wages in preexisting layers.

8.2 Economic intuition

The intuition comes from a knowledge hierarchy. Workers in lower layers solve common problems; managers in higher layers solve less frequent and more difficult problems. When a firm adds a new upper layer, it can economize on knowledge in lower layers because rare problems can be passed farther up the hierarchy. The firm expands, but the average knowledge and wage required in preexisting layers falls.

8.3 Incremental contribution revisited

The paper advances the literature in three main ways. First, it provides a large-scale empirical anatomy of firm hierarchies. Second, it shows that firm growth often involves discrete reorganization rather than smooth scaling. Third, it documents the novel fact that expanding firms can lower wages in preexisting layers when they add managerial layers.

8.4 Implications for firm dynamics and wage inequality

The results imply that models of firm growth should include internal organization. The same increase in value added can have very different wage implications depending on whether the firm reorganizes. This matters for understanding wage inequality within firms, the role of managers, and the organizational response to demand or productivity shocks.

8.5 Limitations

The analysis is descriptive and theory-guided rather than based on randomized organizational shocks. The data are strong at the firm-layer level but weaker for following every individual worker through the reorganization process. The adjustment mechanisms may also depend on French labor market institutions. Finally, the mapping from occupations to layers is empirically validated but still an approximation to the true internal organization of firms.

8.6 Takeaway

The enduring takeaway is that firm organization is not a black box. Firms have measurable hierarchies, and those hierarchies change as firms grow. Reorganization through layers is a central

margin of adjustment, and it can reverse the usual positive relationship between firm growth and wages in existing parts of the firm.